



University of Piraeus

School of Information and Telecommunications Technologies

Department of Digital Systems

Level: Postgraduate Program of Study

Security of Wireless and Mobile Networks

Case 1

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Abstract

The understanding of the structure of the IEEE 802.11 protocol was achieved through wireless traffic monitoring, after the network card was successfully set to Monitor Mode allowing the capture of raw frames. Then, by leveraging the Wireshark tool and applying targeted view filters, it was possible to identify and analyze the protocol's three main categories. This process not only confirmed in practice the theoretical architecture of wireless packets, but also highlighted the absence of encryption and mandatory authentication in management messages, thus identifying the weakness of the template that was the necessary background for the active attack on Use Case 2.

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Introduction

In this exercise, we will focus on recording and analyzing the wireless network traffic (frames) using the Wireshark tool. To achieve this, it is necessary to turn the wireless network card (Wi-Fi adapter) to "Monitor Mode". If it remains in the default "Managed Mode", the operating system will filter the special headers of the 802.11 protocol, showing us only the classic Ethernet traffic.

The IEEE 802.11 standard defines the protocols for implementing wireless local area networks (WLANs), which is what we commercially call Wi-Fi. In the OSI theoretical model, 802.11 operates at the two lowest levels:

- Physical Layer (PHY)
- Data Link Layer (DLL)

Monitoring at this low level is critical. If we were to look at higher-level protocols, such as HTTP (Application Layer), we would only see the "content" of the communication (e.g., which web pages load or what files are downloaded), which is "hidden" within the data messages. HTTP, however, does not reveal how devices communicate to maintain their physical connection. Without the resolution of 802.11, we lose visibility into MAC addresses and network control and management messages.

According to the theory, the 802.11 frames that we will search for and analyze are divided into three main categories:

- **Control Messages:** They facilitate the exchange of data frames between stations. Typical examples are ACK, RTS and CTS messages.
- **Management Messages:** Allow communication to be maintained or interrupted (e.g., beacon frames emitted by routers to indicate their presence and association requests when a device requests to connect). This is where the essence of security lies: these messages are not always authenticated. Although every 802.11 station is required to use the authentication service before establishing a connection, Several "administrative" messages circulate freely. This practically means that, while data (Data Messages) are encrypted, the network does not always strictly control whether the one sending a management frame is actually the one it supports.

- **Data Messages:** These are the frames that carry the actual payload, such as packets from web pages, files, etc.

As a natural continuation of the above analysis, in Use Case 2 we will proceed to the practical exploitation of the weaknesses of the protocol. Specifically, we will carry out a Packet Injection/Deauthentication attack. Taking advantage of the fact that the deauthentication service is just a notification that cannot be rejected and that the management frames are not always authenticated, we will inject fake interrupt packets into the air. Our goal is to trick the network and force the forced disconnection (drop) of a device from the router.

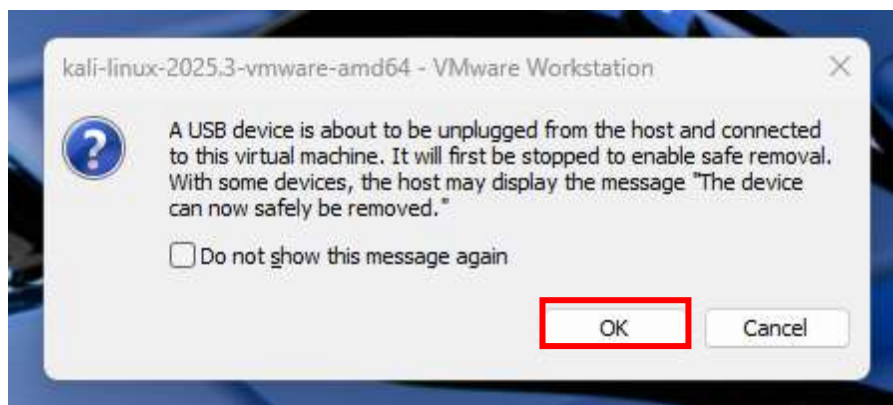
Practical piece of work

Regarding the first step in connecting the equipment to VMware, the process begins with the installation of the wireless USB network adapter **TP-LINK TL-WN722N v4 (150Mbps) with Detachable Antenna** in a free port of the laptop. Then, through the VMware main menu (at the top of the screen), we go to the "VM>Removable Devices" path and locate the specific device in the list, which is usually listed with the name of the manufacturer. Finally, by selecting the "Connect (Disconnect from host)" command, the card is disconnected from the main operating system (Windows) and assigned exclusively for use in the Kali Linux environment, being the main tool for the successful execution of the task.



When the device is connected, the VMware environment displays a standard information message confirming that the TP-Link network card is disengaged from the host machine (Windows) and is attached exclusively to the Kali Linux virtual machine.

order
verify



In
to
the

successful recognition of the wireless interface by the operating system, this is followed by launching a terminal window within Kali Linux and executing the command

iwconfig.

```
(kali@kali)-[~]
$ iwconfig
lo          no wireless extensions.

eth0        no wireless extensions.

br-480f97c7b15a  no wireless extensions.

docker0     no wireless extensions.

wlan0       IEEE 802.11  ESSID:off/any
            Mode:Managed  Access Point: Not-Associated  Tx-Power=20 dBm
            Retry short limit:7  RTS thr=2347 B  Fragment thr:off
            Power Management:off

(kali@kali)-[~]
$
```

As can be seen from the results of the command execution, the Kali Linux operating system successfully recognized the external network adapter, assigning it the interface designation **wlan0**. At the same time, it was found that the default mode of the card was "Managed". In order to switch to error-free Monitor Mode, it was deemed necessary to terminate any system processes and network services (such as NetworkManager) that may cause interference or attempt to arbitrarily restore the card to its original state. To this end, the order was executed **sudo airmon-ng check kill**, thus ensuring seamless preparation of the wireless interface for the next stages of analysis.

```
(kali@kali)-[~]
$ sudo airmon-ng check kill
[sudo] password for kali:

Killing these processes:

    PID Name
    3346 wpa_supplicant

(kali@kali)-[~]
$
```

When executing the previous command, the system successfully detected and terminated the process **wpa_supplicant**, which is responsible for managing standard wireless connections in the background. By ensuring the right environment and eliminating possible interference, the network card transition took place **wlan0** in tracking mode (**Monitor Mode**) through the mandate **sudo airmon-ng start wlan0**. Finally, to definitively confirm the successful change of the operating state of the

interface, the command was re-executed **iwconfig**.

```
(kali@kali)-[~]
$ sudo airmon-ng start wlan0

PHY      Interface  Driver      Chipset
phy0     wlan0      rtl8xxxu    TP-Link TL-WN722N v2/v3 [Realtek RTL8188EUS]
          (monitor mode enabled)

(kali@kali)-[~]
$ iwconfig

lo        no wireless extensions.

eth0      no wireless extensions.

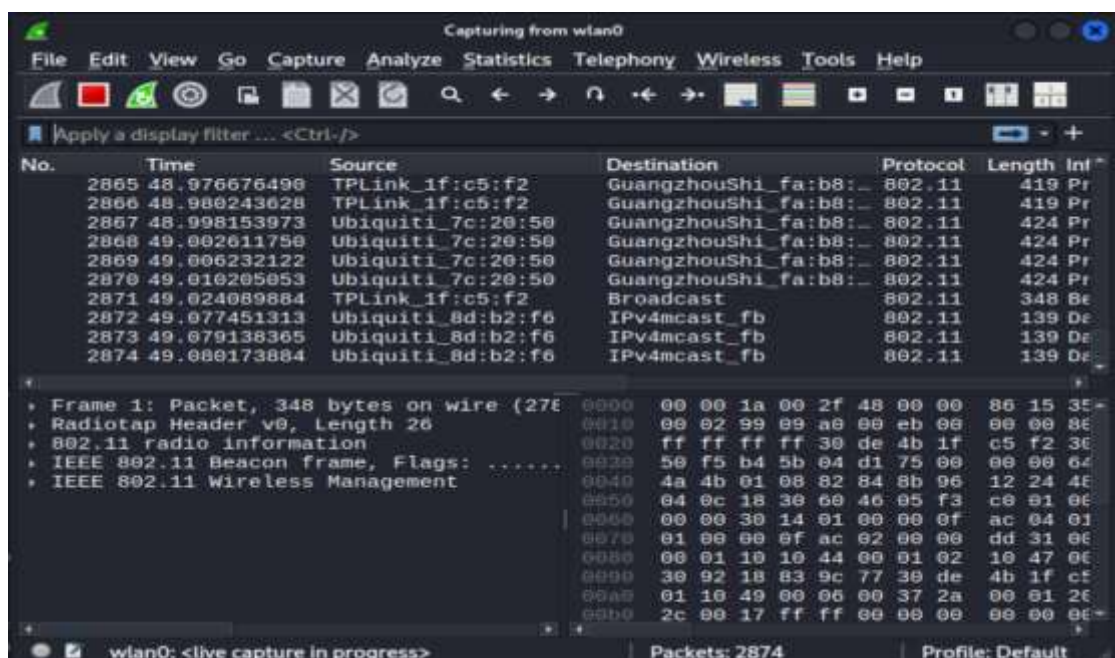
br-480f97c7b15a no wireless extensions.

docker0   no wireless extensions.

wlan0     IEEE 802.11  Mode:Monitor  Frequency:2.457 GHz  Tx-Power=20 dBm
          Retry short limit:7  RTS thr=2347 B  Fragment thr:off
          Power Management:off

(kali@kali)-[~]
$
```

As confirmed by the results of the previous command, the working state of the interface was successfully changed to "Monitor". Having now configured the wireless antenna to intercept network traffic, the process proceeded to the main stage of Use Case 1, which concerns the recording and analysis of wireless frames. To this end, the order was executed **sudo wireshark** in the terminal, by launching the protocol analysis tool of the same name with administrator privileges, in order to ensure full access to the network hardware.



In order to isolate the desired traffic from the overall "noise" of the network and to classify the collected frames into the three main categories of the 802.11 standard, the specialized display filters of the Wireshark tool were used. Since the category of

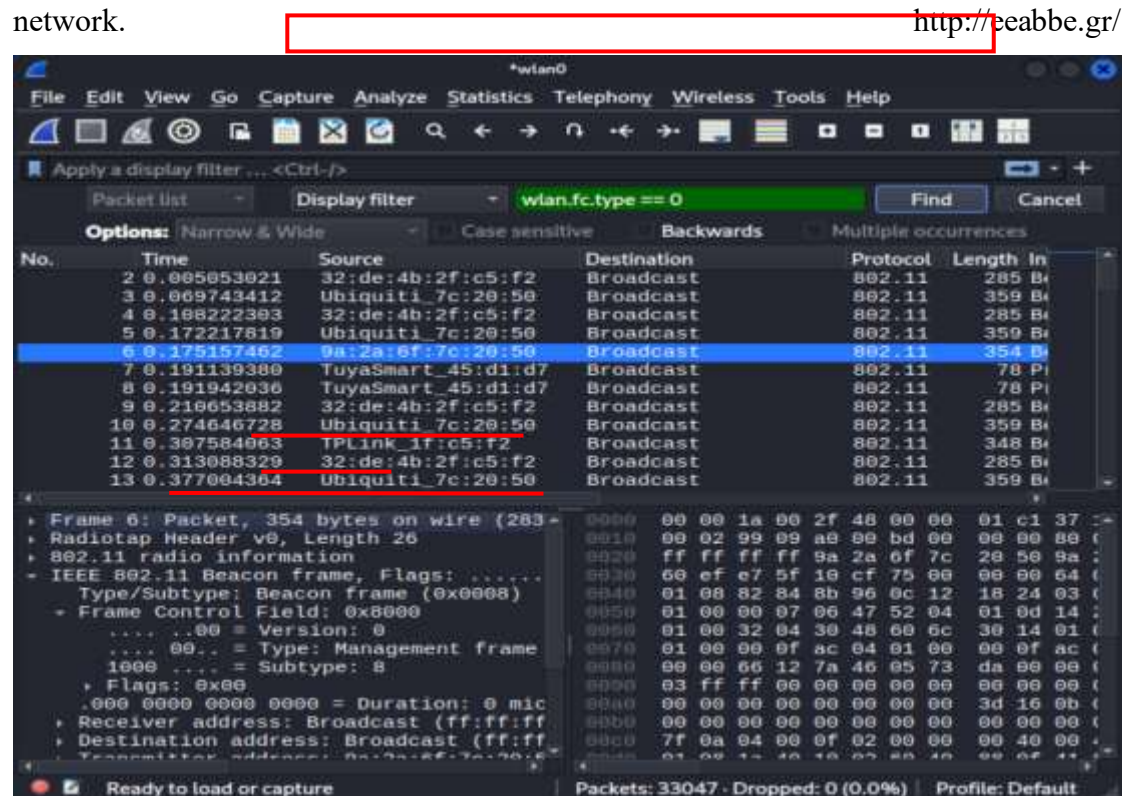
each wireless message is determined solely by the value of the "Type" field within the Frame Control header, the corresponding search parameters were applied sequentially.

First, by using the filter **wlan.fc.type == 0** management frameworks (**Management Frames**), which control the connection and include messages such as Beacons, Probe Requests, and Authentication/Association.

Then, the filter **wlan.fc.type == 1** allowed the control panels to be viewed exclusively (**Control Frames**), which include transmission assistance messages such as Acknowledgement (ACK), RTS and CTS.

Finally, through the filter **wlan.fc.type == 2** data frameworks (**Data Frames**), which bear the actual payload of users.

Applying this analysis methodology, the packet with serial number 6 of the log, which was successfully identified as a Beacon frame (a subcategory of Management Frames), was examined in detail, practically confirming the mechanism through which the router periodically announces its presence and characteristics in the wireless network.



In-depth analysis of the "Frame Control Field" fully verified the theoretical structure of the 802.11 protocol. In particular, the decoding of the individual fields confirmed the following technical characteristics:

- Protocol Version: A value of 0 was recorded, which corresponds to the current version of the protocol.
- Frame Type: A value of 0 (00 in binary) has been detected, which certifies the character of the packet as a Management Frame.
- Subtype: Recorded a value of 8, which identifies this frame as a Beacon message.

In addition to the categorization of the framework, the examination of its routing parameters is of particular interest. As it was established, the MAC universal broadcast address (Broadcast-FF:FF:FF:FF:FF:FF) is used as the destination and receiver address. This finding is absolutely pivotal, as it proves the mechanism of operation of Beacon frames. Specifically, the router (Access Point) does not target a specific recipient, but transmits these messages indiscriminately to all the nodes in its range. This ensures that all nearby devices are immediately informed about the existence of the network, as well as the technical characteristics required to successfully connect to it.

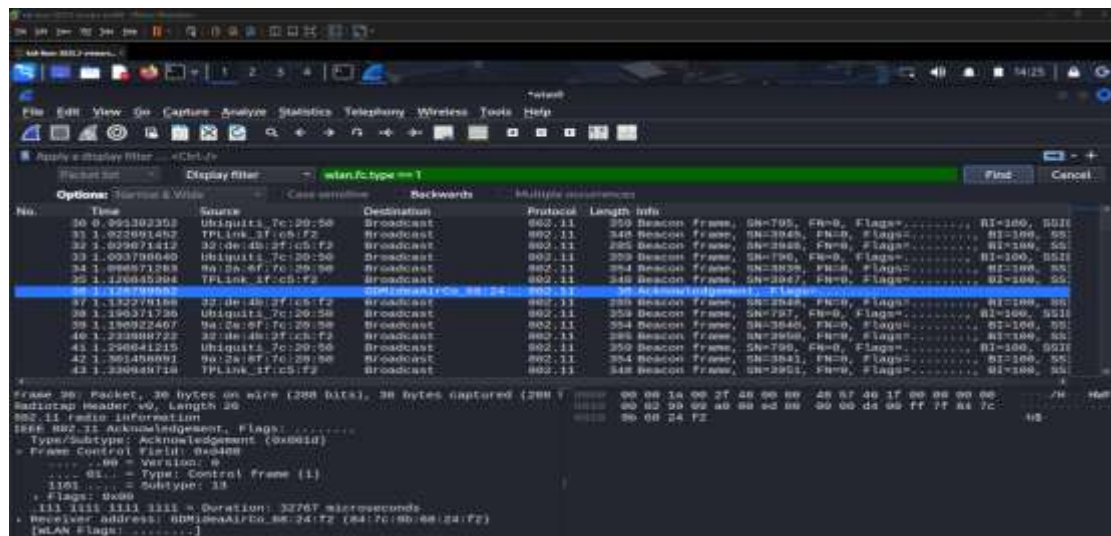
```
Receiver address: Broadcast (ff:ff:ff:ff:ff:ff)
Destination address: Broadcast (ff:ff:ff:ff:ff:ff)
```

With the filter "wlan.fc.type == 1", Wireshark successfully isolated the Control Frames. Analysis of the selected package (No. 36) revealed the following technical characteristics through the decoding of the "Frame Control Field":

- Frame Type: Recorded a value of 1 (01 in binary), certifying that the packet belongs to the category of Control Frames.
- Subtype: A value of 13 was recorded, which, based on the 802.11 standard, corresponds exclusively to Acknowledgement Confirmation Messages (ACK).

These findings are fully in line with the theoretical framework for the operation of wireless networks. According to the theory, the role of Control frames lies in facilitating and reliably exchanging data (data frames) between the stations involved. The detected ACK message stands as the most prominent example of this functionality, as it serves as an explicit confirmation of the successful reception of a data packet,

ensuring the integrity of the communication.

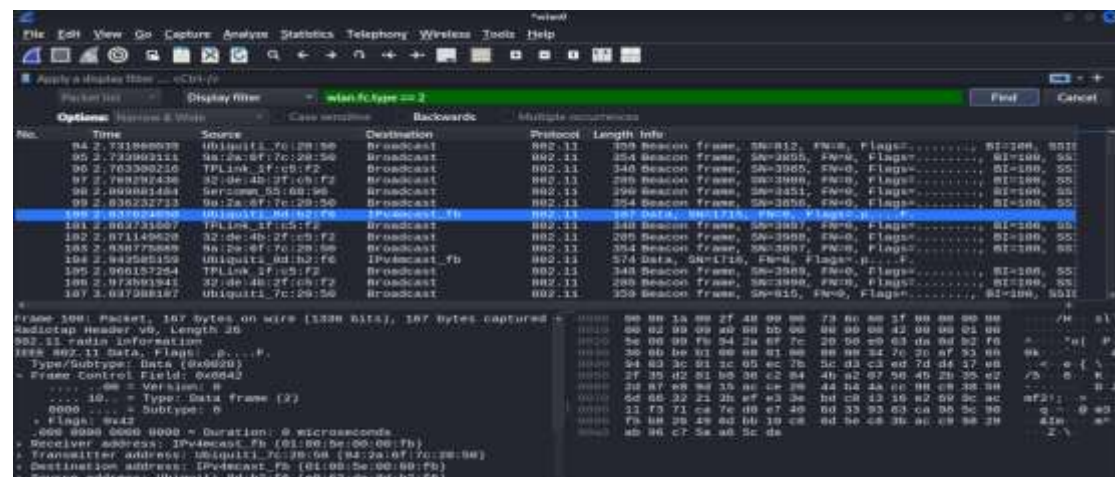


By applying the filter "wlan.fc.type == 2", the Data Frames were successfully isolated. Analysis of the selected package (No. 100) revealed the following technical characteristics when decoding the "Frame Control Field":

- Frame Type: A value of 2 (10 in binary) was recorded, which confirms the inclusion of the packet in the third basic category of the protocol (Data Messages).
- Subtype: A value of 0 was recorded, indicating that it is a standard data transfer frame (Data).

The above findings are completely identical to the theoretical background of the 802.11 standard. As predicted by theory, this category of frames is solely responsible for the transfer of the actual payload (payload) to the network. In practice, this means that through these packets essential user information is trafficked, such as web browsing data (e.g. HTTP traffic), file transfers and the general communication of higher-level

applications.



Conclusion

By completing the first stage of the work, the practical monitoring and analysis of wireless traffic in the Data Link Layer of the IEEE 802.11 standard was achieved. The transition of the wireless network card to Monitor Mode proved to be the decisive step, as it allowed to bypass the limitations of the operating system and gain full visibility into the raw frames exchanged on the wireless medium. Through the use of targeted filters in the Wireshark tool, it was possible to identify, isolate and analyze the Frame Control Field for the protocol's three main message categories:

- Management Frames (Type 0): The existence of Beacon frames (Subtype 8), which are broadcast freely from Access Points to indicate their presence in the network, has been confirmed.
- Control Frames (Type 1): ACK (Subtype 13) confirmation messages have been identified, which are essential for checking smooth and error-free transmission.
- Data Frames (Type 2): Data frames (Subtype 0) have been identified, which are solely responsible for transferring the actual payload of communication.

An in-depth understanding of the structure of these packets is not just a theoretical exercise, but the foundation of offensive security in wireless networks. Practical observation confirmed the inherent weakness of the standard: the absence of mandatory encryption and authentication in critical management messages. It is precisely this finding that is the necessary theoretical and practical springboard for Use Case 2, where the transition from passive monitoring (sniffing) to active exploitation

(packet injection) takes place, through the implementation of a targeted Deauthentication attack.

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